DIOGO CATARINO

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PROFESSIONAL SUMMARY

Software Development Engineer II at Amazon with a Master's in Computer Science and Engineering. Specialized in building full-stack financial and operational platforms, conversational domain AI agents, Model Context Protocol (MCP) tooling, and autonomous multi-agent workflows that drive multi-million dollar operational savings.

EXPERIENCE

Software Development Engineer II Sep 2026 - Present
Amazon *Luxembourg*

- Building a full-stack **AI Finance & ROI Platform** (React, TypeScript, Python, AWS) to track infrastructure cost, resource utilization, and revenue/savings attribution across internal AI projects at scale.
- Developing **Project Logos**, a conversational AI agent engineered with deep supply chain domain knowledge (Strands Agents, MCP) to enable natural-language investigation of operational metrics and fulfillment flows.

Business Intelligence Engineer I Aug 2024 - Sep 2026
Amazon *Luxembourg*

- Architected an autonomous AI Agent swarm (agentic workflow) using Strands and MCP servers to automate the "Volume Loss to Caps" bridge, optimizing human-in-the-loop workflows and saving **109,500 man-hours annually (est. $8.2M+ cost reduction)**.
- Developed a centralized Goal Tracking platform (React, TypeScript, DynamoDB) for 130+ users, automating 500+ Director-level targets and reducing cross-team alignment latency by **70%**.
- Built an autonomous AI agent using Strands and MCP servers to retrieve metrics and automatically generate weekly WBR narrative bridges, saving **5+ hours per week** for L6 leadership, alongside a VP-sponsored KPI management engine.

Software Development Engineer - Performance Science Feb 2024 - Aug 2024
The Goodyear Tire & Rubber Company *Luxembourg*

- Built a full-stack Lateral Force (LDF) testing visualization tool (React.js, Redux Toolkit, FastAPI) integrating ANOVA, post-hoc tests, and Bayesian inference to automate tire performance comparisons and surface R&D insights.
- Designed and implemented robust data pipelines and query APIs at scale (AWS Athena, SQL) to process, transform, and analyze global tire sensor datasets for performance science teams.

PROJECTS

- **Wreck Atlas** (shipwreckatlas.com ◇ github.com/diocata/wreck-atlas) — Map-first web application charting 100,000+ documented shipwrecks and underwater obstructions worldwide with an interactive sonar viewport, spatial indexing, and a public OpenAPI 3.1 REST API (TypeScript, Zustand).
- **Discord Bot MCP Server** (github.com/diocata/discord-bot-mcp) — Safety-scoped Model Context Protocol (MCP) server exposing Discord Bot REST API operations as permission-bounded tools for AI assistants like Claude and Cursor with explicit channel allowlists and rate-limit protection (TypeScript, Node.js).

SKILLS

Languages	Python, TypeScript, JavaScript, SQL, Java, Scala, R
Full-Stack & APIs	React, Next.js, Redux, Zustand, Tailwind CSS, FastAPI, Node.js, OpenAPI 3.1, D3.js
Cloud & Data	AWS (Lambda, API Gateway, DynamoDB, Athena, S3, SES), PyTorch, Pandas, Computer Vision, ANOVA, Docker
Agentic & AI	Strands Agents, Model Context Protocol (MCP), AWS AgentCore, Conversational AI Agents, LLMs

EDUCATION

Master of Computer Science and Engineering, Instituto Superior Técnico 2022 - 2024
- Relevant Coursework: Deep Learning, Data Analysis & Integration, Decision Support Models, Information Visualization.
Master Thesis: Recognising Portuguese Stones with Deep Learning.
- Designed and trained Computer Vision models using PyTorch; preprocessed and curated image datasets.

Erasmus, Computer Science & Data Science, University of Würzburg 2023
- Relevant Coursework: Topics in Data Science, Data Science.

Bachelor of Computer Science and Engineering, ISCTE - IUL 2019 - 2022